Swarm Velocity Dynamics in US Sector ETFs *

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Abstract

This study examines whether the leader-following swarm velocity of Chen et al. (2025), estimated for eleven Select Sector SPDR exchange-traded funds (ETFs) at daily frequency, carries tradable information. We backtest two strategies: a threshold rule that applies the pooled velocity to a leader-versus-loser pair trade, and a cross-sectional rule that ranks the ETFs by their own velocities. The velocity threshold cuts transaction costs but does not select better trades out of sample, and neither rule earns a significantly positive return. The estimated speeds are negative and lie close to the value that returns without leader-following produce, because the gap to the leader is dated on the same day as the velocity. An equally weighted portfolio of the same ETFs has a higher Sharpe ratio than both cross-sectional portfolios.[1]

1 Introduction

A swarm is a population of simple agents whose collective motion emerges from local rules rather than central direction. Reynolds (1987) simulated flocks, herds and schools whose aggregate motion emerges from three local rules applied by each agent, namely collision avoidance, velocity matching and flock centering. Eberhart and Kennedy (1995) adapted the same idea into particle swarm optimization, in which each particle accelerates toward the best position it has found and toward the best position found by the swarm, so that the population searches for an optimum without gradients.

The financial interest lies in the leader term. Herding and lead-lag effects are documented features of asset returns, and momentum, which buys recent winners and sells recent losers, has been one of the most persistent anomalies since Jegadeesh and Titman (1993). A swarm model gives these phenomena a quantitative handle, since the coefficient on the leader term measures how quickly a follower closes the gap to the leader. Chen et al. (2025) estimate this speed for 190 risk factors at monthly frequency, sort the factors into quintiles, and report that slow-moving factors earn 0.26% per month more than fast-moving ones, a spread they attribute to delayed information processing.

This study extends the leader-following swarm velocity estimator of Chen et al. (2025) to eleven Select Sector SPDR ETFs at daily frequency. To assess whether the estimated velocity carries tradable information, we backtest a threshold rule that applies the pooled velocity to a leader-versus-loser pair trade and a cross-sectional rule that ranks the ETFs by their own velocities. The velocity threshold cuts transaction costs but leaves the trade unprofitable out of sample, and the fast-minus-slow portfolio earns +12.0%, without statistical significance, and trails an equally weighted portfolio of the same ETFs. Section 2 describes the data. Section 3 states the swarm velocity estimator. Section 4 specifies the threshold rule, and Section 5 specifies the cross-sectional rule. Section 6 concludes.

2 Data

This study focuses on the universe of eleven Select Sector SPDR ETFs. They together partition the S&P 500 into materials, communication services, energy, financials, industrials, technology, consumer staples, real estate, utilities, health care and consumer discretionary. Daily adjusted

*This is a class project for Fordham University's QFGB 890P course.

[1]Code and replication data are available at *GitHub*.

closing prices were collected from October 2018 to December 2025, and $r_{i,t}$ denotes the daily log return of ETF i on day t. The training sample runs from October 2018 to December 2023, and the testing sample runs from January 2024 to December 2025. Appendix B reports summary statistics.

3 Swarm Velocity

Each ETF is embedded in an L-dimensional space whose coordinates are its lagged returns, and the empirical velocity is the change in position between consecutive days,

$$\boldsymbol{x}_{i,t} = (r_{i,t-1}, \ldots, r_{i,t-L}), \qquad \boldsymbol{\nu}_{i,t} = \boldsymbol{x}_{i,t} - \boldsymbol{x}_{i,t-1}, \qquad \boldsymbol{d}_{i,t} = \boldsymbol{x}_{\ell_t,t} - \boldsymbol{x}_{i,t}, \tag{1}$$

with $L = 5$ and ℓ_t the leader. The leader-following model states that velocity is proportional to the gap, $\boldsymbol{\nu}_{i,t} = \alpha \boldsymbol{d}_{i,t} + \boldsymbol{\varepsilon}_{i,t}$, and the least squares estimate for a single ETF, together with the estimate obtained by imposing one common speed on the whole swarm, is

$$\hat{\alpha}_t^{(i)} = \frac{\langle \boldsymbol{\nu}_{i,t}, \boldsymbol{d}_{i,t} \rangle}{\|\boldsymbol{d}_{i,t}\|^2}, \qquad \hat{\alpha}_t = \frac{\sum_i \langle \boldsymbol{\nu}_{i,t}, \boldsymbol{d}_{i,t} \rangle}{\sum_i \|\boldsymbol{d}_{i,t}\|^2}. \tag{2}$$

Appendix A includes the derivation and demonstrates that $\hat{\alpha}_t$ is the average of $\{\hat{\alpha}_t^{(i)}\}$ weighted by $\|\boldsymbol{d}_{i,t}\|^2$. The leader is identified as the ETF with the highest cumulative return over the position window. It satisfies $\boldsymbol{d}_{\ell_t,t} = \mathbf{0}$ and is excluded.

On the training sample, $\hat{\alpha}_t^{(i)}$ has mean -0.388 with standard deviation 0.814 and $\hat{\alpha}_t$ has mean -0.468. Both are negative, whereas Chen et al. (2025) report that every factor's average speed over their sample is positive, with a mode slightly below 0.5. The difference stems from the dating of the gap. The estimator follows Eq. (14) of Chen et al. (2025), which dates the gap at t, whereas their Eq. (13) dates it at $t-1$. Since $\boldsymbol{\nu}_{i,t}$ and $\boldsymbol{d}_{i,t}$ both contain $\boldsymbol{x}_{i,t}$, the numerator carries $-\|\boldsymbol{x}_{i,t}\|^2$, and for serially uncorrelated returns with a common variance and a fixed leader, the ratio of the expected numerator to the expected denominator is $-1/2$ whatever the pairwise correlation, while the gap at $t-1$ yields $+1/2$. Simulated returns without leader-following give a mean pooled estimate between -0.46 and -0.37 for pairwise correlations from 0 to 0.9, and on the training sample the gap at $t-1$ yields $+0.442$. Both estimates therefore lie close to the values that returns without leader-following produce.

4 Threshold Rule

To examine whether the swarm velocity improves a leader-versus-loser trade, we backtest a threshold rule. The threshold rule trades a pair between the day's leader $b_t = \arg\max_i r_{i,t}$ and the day's loser $s_t = \arg\min_i r_{i,t}$, and uses $\tilde{\alpha}_t$, the pooled estimate in (2) with b_t in place of ℓ_t, to decide whether to trade at all. With $\boldsymbol{e}_i$ the ith unit vector and thresholds $\tau_l \le \tau_u$,

$$\boldsymbol{w}_t = \tfrac{1}{2} \begin{cases} +\boldsymbol{e}_{b_t} - \boldsymbol{e}_{s_t}, & \tilde{\alpha}_t > \tau_u \text{ and } r_{b_t,t} > 0, \\ -\boldsymbol{e}_{b_t} + \boldsymbol{e}_{s_t}, & \tilde{\alpha}_t < \tau_l \text{ and } r_{b_t,t} < 0, \\ \mathbf{0} & \text{otherwise}, \end{cases} \tag{3}$$

Half the capital is long, and half is short, with no leverage. The portfolio is rebalanced daily. Positions are set at the close of day t and earn the return of day $t+1$. Thresholds maximize the total return on the training sample over the grid $\tau_l \in \{-1.3, -1.2, \ldots, -0.4\}$ and $\tau_u \in \{-0.4, -0.3, \ldots, 1.5\}$.

To isolate the contribution of the velocity metric, we benchmark the rule against the same pair trade without the velocity condition. It buys the leader and sells the loser when the leader's

return is positive, and sells the leader and buys the loser when it is negative. The backtest is conducted under a transaction cost of 5 basis points per unit of one-way turnover.

The first two rows of Table 1 compare the rule with its benchmark. The benchmark loses 67.3% of capital in training and 45.8% in testing. In training, it holds a position on 99.8% of days, and turnover costs 21.8% of capital a year. The threshold rule is active on 12.0% of training days and 6.7% of testing days, which cuts the annual cost to 2.8% and 1.5% of capital. The smaller loss reflects this lower activity rather than better trades. In training, where the thresholds are fitted, the gated trades earn +0.11% per active day before costs against +0.01% for the benchmark, but in testing they earn −0.06% against −0.03%.

The threshold rule returns +2.7% in training at a Sharpe ratio of +0.12, the best of the 200 cells, but −5.0% in testing at −0.76. Figure 1 confirms that this is not a matter of threshold choice: only 16 of 200 training cells and 2 of 200 testing cells are profitable, and the best testing cell returns +0.3%.

	Training			Testing		
	Return	Volatility	Sharpe	Return	Volatility	Sharpe
Threshold rule at (−0.5, 0.2)	+2.7%	5.7%	+0.12	−5.0%	3.3%	−0.76
Benchmark	−67.3%	14.9%	−1.37	−45.8%	11.1%	−2.75
Opposite rule	−28.3%	5.7%	−1.09	−1.0%	3.4%	−0.14
Opposite benchmark	−72.1%	14.9%	−1.57	−24.7%	11.1%	−1.25

Table 1: Total return, annualized volatility and annualized Sharpe ratio at a zero risk-free rate, net of transaction costs.

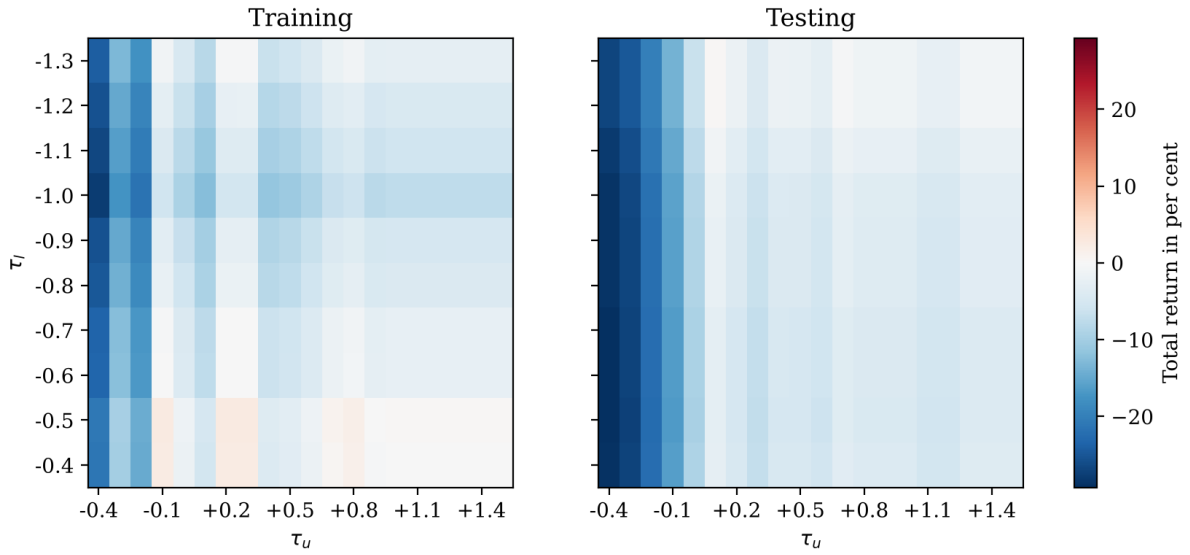


Figure 1: Total return of the threshold rule over the grid of thresholds.

Since the trade loses under almost every calibration, we ask whether its direction is simply inverted. We reverse the long-short position and the results are shown in the lower block of Table 1 and, for the testing sample, in Figure 2. The answer is negative. Reversing the benchmark shows no improvement in training at −72.1%. The opposite rule loses 28.3% in training and 1.0% in testing, and neither testing return is significant ($t = -1.06$ and -0.20).

Therefore, the velocity threshold cuts transaction costs but does not select better trades out of sample.

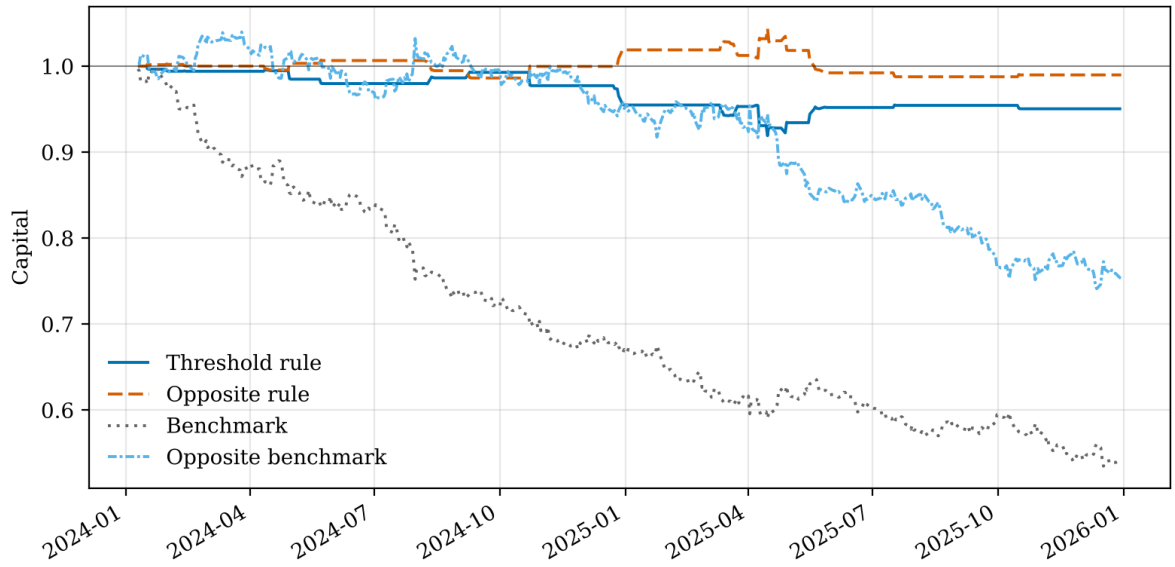


Figure 2: Capital of the threshold rule, the opposite rule and both benchmarks over the testing sample.

5 Cross-Sectional Rule

The cross-sectional rule follows the design of Chen et al. (2025). On each formation day, the ETFs are ranked on $\hat{\alpha}_t^{(i)}$, and the slow-minus-fast portfolio buys the k slowest and sells the k fastest in equal weights,

$$\boldsymbol{w}_t^{(1)} = \frac{1}{2k}\left(\sum_{i\in\mathcal{L}_t} \boldsymbol{e}_i - \sum_{i\in\mathcal{H}_t} \boldsymbol{e}_i\right). \tag{4}$$

Conversely, the fast-minus-slow portfolio sells the k slowest and buys the k fastest in equal weights,

$$\boldsymbol{w}_t^{(2)} = \frac{1}{2k}\left(\sum_{i\in\mathcal{H}_t} \boldsymbol{e}_i - \sum_{i\in\mathcal{L}_t} \boldsymbol{e}_i\right), \tag{5}$$

where $\mathcal{L}_t$ and $\mathcal{H}_t$ collect the k lowest and k highest estimates, so that half the capital is long and half is short. We set $k = 2$, approximating the quintile breadth of Chen et al. (2025) on each side. The portfolio is rebalanced every h trading days, and the baseline is set to a one-month horizon, $h = 21$.

Figure 3 shows the backtest performance. The slow-minus-fast strategy returns -25.7% over the full sample, October 2018 to December 2025, at a Sharpe ratio of -0.48, and the fast-minus-slow strategy returns $+12.0\%$ at a Sharpe ratio of $+0.24$. We benchmark these strategies with an equally weighted portfolio of the eleven ETFs rebalanced on the same dates. The equally weighted benchmark returns $+131.1\%$ at a Sharpe ratio of $+0.81$, higher than both rules, although it is long only and carries the market exposure that the dollar-neutral rules avoid.

To test the robustness of the holding period h, Table 2 and Figure 4 vary h around that baseline over $\{1, 2, 3, 5, 10, 21, 63\}$. The slow-minus-fast strategy has negative performance for all holding periods, while the fast-minus-slow strategy yields positive returns when h is at least 10. First, a shorter h implies higher turnover and hence higher transaction costs. Second, the mean cross-sectional rank correlation between $\hat{\alpha}_t^{(i)}$ and the subsequent h-day return rises from

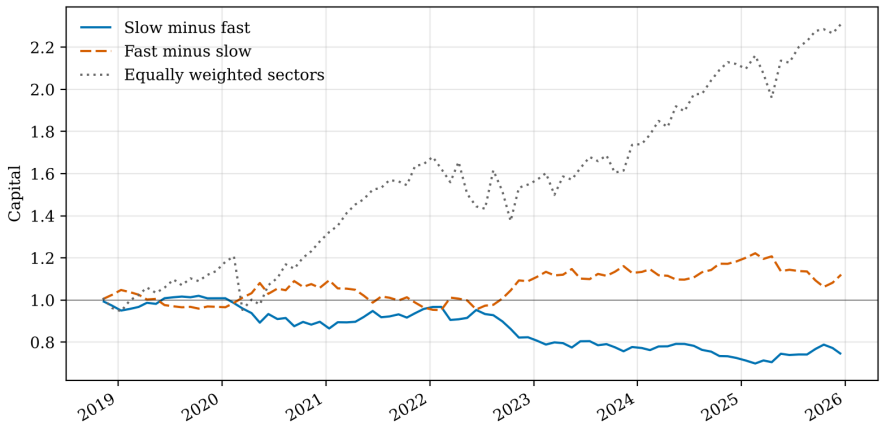


Figure 3: Capital of the cross-sectional rule, against an equally weighted portfolio.

+0.008 at 5 days to +0.016 at 10 days, +0.018 at 21 days and +0.071 at 63 days, although none differs significantly from zero (largest $t = 1.11$).

	Slow minus fast		Fast minus slow	
h	Return	Sharpe	Return	Sharpe
1	−69.8%	−2.11	−51.0%	−1.24
2	−62.5%	−1.66	−19.5%	−0.34
3	−52.0%	−1.29	−16.0%	−0.28
5	−38.3%	−0.80	−14.4%	−0.23
10	−32.7%	−0.65	+7.5%	+0.16
21	−25.7%	−0.48	+12.0%	+0.24
63	−17.8%	−0.35	+12.4%	+0.27

Table 2: Total return and Sharpe ratio of the cross-sectional rule by holding period h in trading days, net of transaction costs and with the Sharpe ratio at a zero risk-free rate. Annualized volatility lies between 7.2% and 8.0% in every cell.

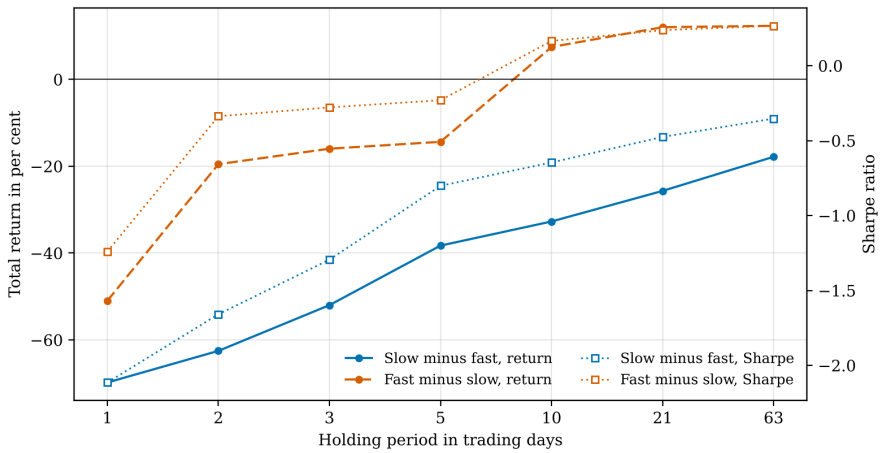


Figure 4: Total return and Sharpe ratio of the cross-sectional rule against the holding period.

The positive rank correlation is opposite in sign to the negative high-minus-low spread that Chen et al. (2025) report for 190 factors, but it is not significant at any horizon, and neither is

the return of the fast-minus-slow portfolio ($t = 0.63$ at $h = 21$). Since the speed estimate lies close to its value without leader-following (Section 3), the sort carries little information on how quickly an ETF closes in on the leader.

6 Conclusion

This study examines whether the swarm velocity of Chen et al. (2025) carries tradable information for eleven sector ETFs at daily frequency, using a threshold rule and a cross-sectional rule. The speed estimates are negative because the gap to the leader is dated on the same day as the velocity, and they lie close to the value that returns without leader-following produce.

The velocity threshold reduces turnover costs but does not select better trades out of sample. The fast-minus-slow portfolio earns positive returns for holding periods of 10 days or more, +12.0% at a Sharpe ratio of +0.24 with a monthly holding period, but not significantly, and it trails an equally weighted portfolio of the same ETFs, which returns +131.1% at a Sharpe ratio of +0.81.

The evidence is limited by the eleven-asset universe and the 502-day testing sample. Future research could date the gap at $t - 1$, as in Eq. (13) of Chen et al. (2025), and test the speed against its value without leader-following.

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A Closed Form for the Velocity Estimate

Imposing one common speed on the swarm, the sum of squared errors at date t is

$$S(\alpha) = \sum_i \|\boldsymbol{\nu}_{i,t} - \alpha \boldsymbol{d}_{i,t}\|^2$$

$$S'(\alpha) = -2\sum_i \langle \boldsymbol{\nu}_{i,t}, \boldsymbol{d}_{i,t} \rangle + 2\alpha \sum_i \|\boldsymbol{d}_{i,t}\|^2, \qquad S''(\alpha) = 2\sum_i \|\boldsymbol{d}_{i,t}\|^2,$$

so S is strictly convex whenever at least one gap is non-zero, and its unique minimizer is the second expression in (2). Restricting the sum to a single term yields the first. The two are related by

$$\hat{\alpha}_t = \sum_i \frac{\|\boldsymbol{d}_{i,t}\|^2}{\sum_j \|\boldsymbol{d}_{j,t}\|^2} \hat{\alpha}_t^{(i)},$$

so the pooled estimate weights each ETF by the squared length of its own gap.

B Summary Statistics

ETF	Mean log return (%)	Volatility (%)	Skewness	Excess kurtosis
XLB	8.1	22.5	−0.41	9.6
XLC	13.1	22.9	−0.48	7.1
XLE	6.6	33.1	−0.94	14.8
XLF	11.4	24.2	−0.58	15.1
XLI	11.0	21.9	−0.51	13.0
XLK	19.4	27.0	−0.27	8.3
XLP	7.8	15.9	−0.46	15.5
XLRE	6.4	22.7	−1.04	17.1
XLU	9.9	21.0	−0.25	15.1
XLV	8.4	17.8	−0.40	10.1
XLY	10.8	24.7	−0.53	7.4

Table 3: Summary statistics of daily log returns, with mean and volatility annualized. Every sector is left skewed and heavy tailed.

	XLB	XLC	XLE	XLF	XLI	XLK	XLP	XLRE	XLU	XLV	XLY
XLB	1.00										
XLC	0.68	1.00									
XLE	0.67	0.45	1.00								
XLF	0.84	0.68	0.71	1.00							
XLI	0.89	0.70	0.69	0.89	1.00						
XLK	0.70	0.83	0.45	0.67	0.75	1.00					
XLP	0.66	0.56	0.42	0.65	0.66	0.54	1.00				
XLRE	0.71	0.60	0.48	0.69	0.72	0.59	0.71	1.00			
XLU	0.59	0.45	0.40	0.58	0.60	0.45	0.73	0.76	1.00		
XLV	0.71	0.64	0.47	0.69	0.72	0.66	0.74	0.69	0.63	1.00	
XLY	0.75	0.82	0.49	0.73	0.78	0.84	0.57	0.65	0.46	0.63	1.00

Table 4: Correlation matrix. The mean off-diagonal entry is 0.65, and the range is 0.40 to 0.89.